IoT based Vertical Lift Module

Meen Patel
Department of Electronics and Communication Nirma University
Ahmedabad, India
21bec091@nirmauni.ac.in

***Abstract*—Vertical Lift Modules (VLMs) are a breakthrough in warehouse automation, providing compact, space-efficient storage solutions that enhance picking efficiency and worker ergonomics. This project details the design and development of a VLM system with control through Internet of Things. The system organizes items on vertically moving trays, optimizing storage by minimizing the required floor space. Allowing remote controlling and monitoring, our VLM setup enables swift and effective item retrieval, streamlining operations and reducing worker movement.**

***Keywords*—Vertical Lift Module, Arduino Nano, Warehouse Automation, Internet of Things, Client-Server Model.**

I. INTRODUCTION

A Vertical Lift Module (VLM) is a vertical automated storage and retrieval system (AS/RS). It comprises trays in the front and back and an inserter/extractor system that runs down the centre. Trays are automatically retrieved and delivered to a pick window for picking. When complete the inserter/extractor will retrieve and store the tray in either the designated position or automatically assign a position based on using the least amount of space or fastest/slowest accessibility (all user-defined). Trays can be stored based on utilization, size, weight, or another custom parameter. Its modular design and construction make it easy to modify and change heights, and locations and pick windows anytime.

The Internet of Things (IoT) is a transformative technology that connects physical devices to the internet, enabling data collection, analysis, and interaction in real time. Through

IoT, everyday objects can communicate and cooperate with each other and with users, driving greater automation and intelligent decision-making across various industries. With applications spanning from smart homes to industrial automation, IoT facilitates enhanced monitoring, control, and efficiency. By integrating sensors, microcontrollers, and wireless communication, IoT systems generate valuable insights that improve processes, reduce costs, and increase productivity. In this project, we leverage IoT technology to develop a Vertical Lift Module (VLM) that automates and streamlines the storage and retrieval process, demonstrating how IoT can transform traditional operations.

II. WORKING OF MODEL

The Vertical Lift Module (VLM) system begins with the tray positioned on the floor level for easy access by the user. The VLM accepts two primary inputs: 1) the floor number, and 2) the side of the module. Upon receiving the inputs, the extractor moves to the specified floor, retrieves the selected tray, and transports it to the designated access point. After the user completes their tasks, the system waits for further instructions to return the tray. When prompted, the extractor moves the tray back to its original location, completing one full cycle. In subsequent operations, the extractor will start from the last accessed floor, ensuring efficient and seamless access for the user.

The IoT-based Vertical Lift Module (VLM) connects to a Wi-Fi network and communicates with a Google Sheets API via HTTPS to store and retrieve operational data.

ESP8266 libraries were used to handle Wi-Fi connectivity, HTTPS requests, and secure client communication for the ESP8266 microcontroller. Several pins and variables are defined to control and track the VLM's status. There are pins to relate to the RFID reader's data handling. Wi-Fi Setup is made on ESP8266 by connecting to a specified Wi-Fi network using the Wi-Fi SSID and password. The setup function for connection logic initializes serial communication and attempts to connect to the specified Wi-Fi network.

URL Handling and HTTPS Requests is used for implementing control with IoT. Google Sheets API URL is used for taking and storing data. Data from the VLM is sent and retrieved from a Google Sheets API endpoint, allowing real-time data logging. A Read URL function forms and sends an HTTPS GET request to the Google Sheets API, which reads or writes data depending on the specified status argument. If status is 'read', it retrieves values for Floor and

Side to determine the target location for the VLM. If an error occurs in the request, an error message is printed.

Movement and Floor Navigation Logic Function manages the movement between floors by calculating the difference between the current and target floors. Uses a delay proportional to the distance between floors, simulating the VLM's vertical travel time. The main loop has Status Check and Navigation. The main loop checks for Wi-Fi connection. If connected, the system reads target floor and side data from the API. If Floor is not zero, the module displays the current floor, calls target floor function to move the VLM to the desired floor and side. Updates current floor value to reflect the new current floor. A Display Control Function Controls a seven-segment display to show the current floor by setting specific pins to represent digits 0–9. Error Handling and Indicator is done by LED Blinking. When Wi-Fi is not connected, the loop flashes an LED on pin 2 as an error indicator.

The code provides an interface to control the VLM's vertical movement between floors, manages Wi-Fi and HTTPS communications with a Google Sheets API, and displays the current floor on a seven-segment display. It enables a user to request a specific floor, move the VLM, and track and update its position in real time.

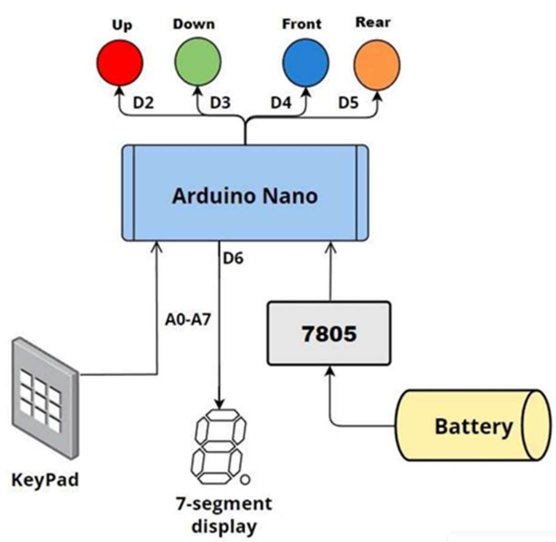

Fig 1. Block diagram of VLM module

III. SYSTEM COMPONENTS

A VLM module consists of a microcontroller, a keypad as an input, 7-segment display, motors, a few LEDs, registers, and a power supply.

A. Microcontroller

A Microcontroller plays a crucial role as the central processing unit or “brain” of the system. Arduino Nano is used here for this purpose. The microcontroller interfaces with all the components, collects input, processes, and provides the output accordingly.

B. Keypad

We have interfaced the 4x4 matrix keypad with Arduino nano board in order to take inputs from the user. Users can select the floor number and rear-front options through the keypad.

C. Output

1. 7- segment display: it will show the current floor number of the tray.

2. LEDs: 4 LEDS are used to notify about the directions up, down, front, and rear.

D. Power Supply

1. A 9V volt battery is utilized to power the system. This makes it low-cost.

2. Voltage Regulator, 7805 is used to bring down 9V from battery to 5V, to supply to the microcontroller.

E. Registers

1K-ohm-8 register (7-segment pins)

1K-ohm-1 register (for LEDs)

IV. IMPLEMENTAION

The implementation of the IoT-based Vertical Lift Module (VLM) involved the integration of hardware and software components to achieve automated storage and retrieval. This system leverages an ESP8266 microcontroller to handle Wi-Fi connectivity and HTTPS communication, Arduino code to control the VLM mechanics, and a Google Sheets API for remote data logging and status monitoring.

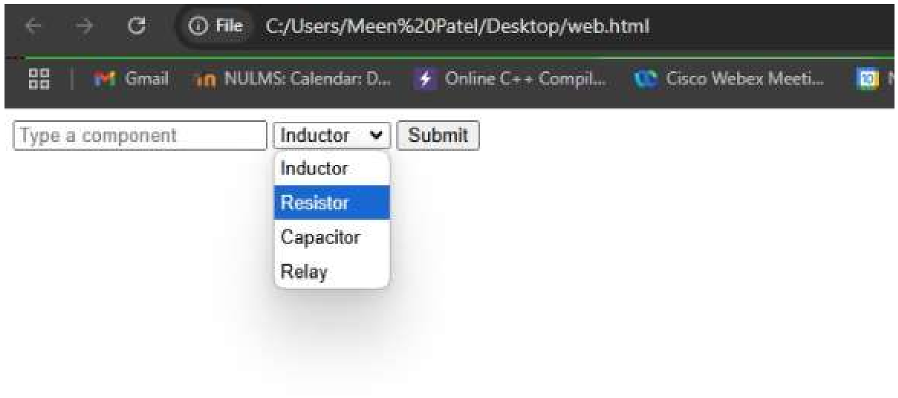

Fig 2. IoT Webpage

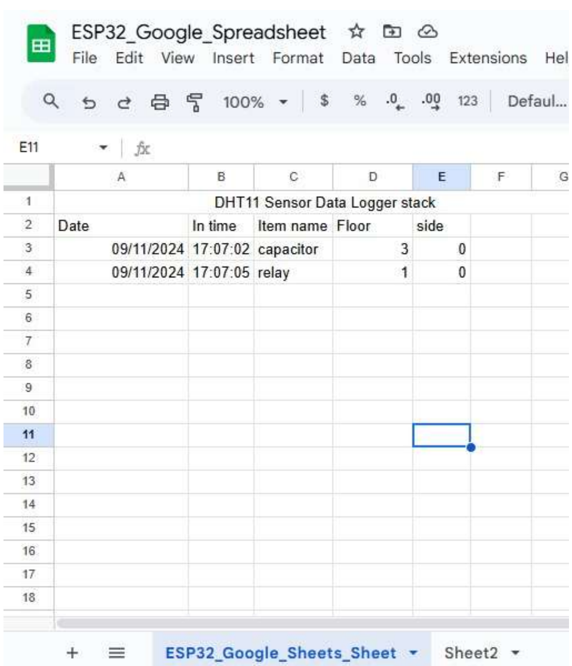

Fig 3. Queue

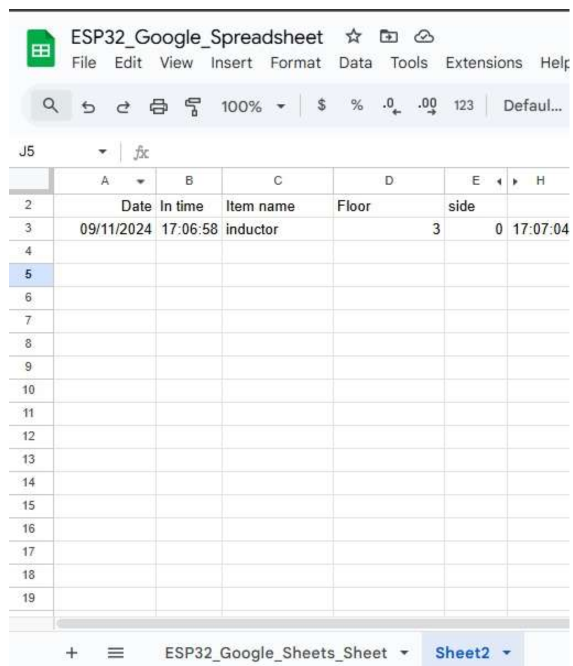

Fig 4. Final Data

In this IoT-based VLM system, user inputs are managed through a web interface, where the user selects the desired floor and side for tray retrieval. These selections are then recorded in a Google Sheet via an API, creating a centralized data source accessible by the microcontroller. The ESP8266 microcontroller, programmed to periodically check for updates, reads this data from the Google Sheet using an HTTPS GET request. This real-time data exchange ensures that the microcontroller always receives the latest commands for VLM operation.

After executing the movement and positioning task, the system updates the Google Sheet to confirm the task's completion and reset the floor position. This bi-directional communication between the microcontroller and Google Sheets enables efficient data logging and status tracking, enhancing system reliability and user experience by providing real-time feedback.

A key feature is that the status of what the VLM is doing at any moment is indicated by the inbuilt LED in the microcontroller. When the inbuilt LED is blinking continuously, it indicates that the microcontroller is not connected to Wi-Fi. When the light is continuously on, the module is processing the current request. The light is continuously off, the module is in rest condition and the seven-segment display will show the current floor.

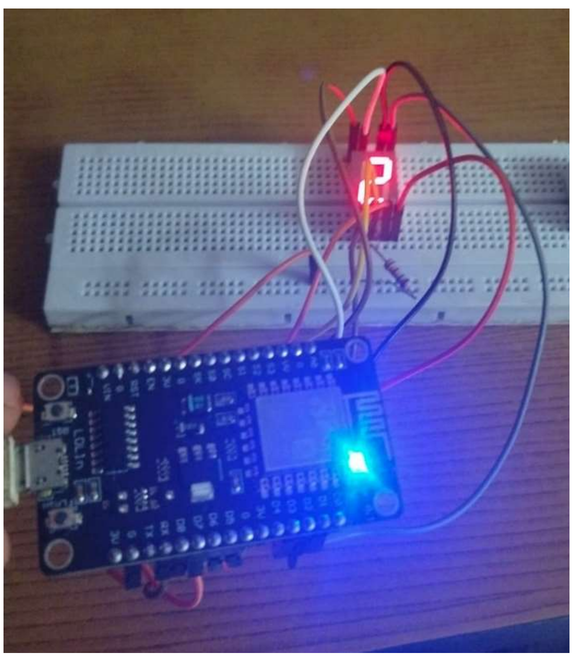

Fig 5. Processing indicated by steady glowing light

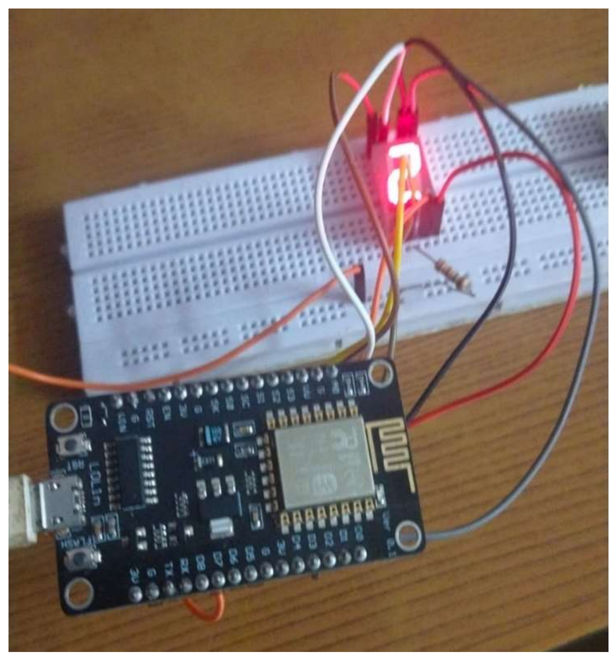

Fig 6. Module in rest condition. Seven segment display showing current floor.

V. APPLICATIONS

Vertical Lift Modules (VLMs) provide significant advantages in space utilization, labour efficiency, and operational effectiveness, making them suitable for diverse applications across various industries.

- Warehousing and Distribution Centres
- Manufacturing Facilities
- Retail and E-commerce Fulfilment Centres
- Automotive Industry
- Pharmaceutical and Healthcare Facilities
- Aerospace and Defence Sector
- Document and File Management

VI. CONCLUSION

This report Vertical Lift Modules (VLMs) provide substantial benefits in terms of time savings, labour efficiency, and overall warehouse productivity by optimizing space utilization. By automating the storage and retrieval processes, VLMs streamline operations and enhance order fulfilment, making them an invaluable tool for industries with high-density storage needs.

However, it is crucial to consider factors such as the initial cost, space requirements for the machine, and potential integration challenges with existing warehouse management systems. Despite these considerations, VLMs offer a transformative solution for facilities looking to improve efficiency, reduce labour costs, and maximize storage capacity. For operations focused on efficient order fulfilment, VLMs can prove to be a game-changer in warehouse automation.

VII. REFERENCES

[1] Midas Metcons Pvt. Ltd. "Vertical Lift Module (VLM) Overview." YouTube, https://www.youtube.com/@midasmetconsprivatelimited4233/videos.

[2] Vidir Solutions. "Vertical Lift Module (VLM) Systems." Vidir Solutions, https://vidirsolutions.com/products/vlm.

[3] Kumar, S., & Pandey, A. (2020). "Vertical Lift Module (VLM) for Warehouse Automation." *International Journal of Advanced Research in Engineering and Technology (IJARET)*, 11(6), 25-30.

[4] "Internet of Things (IoT) in Industry 4.0." (2020). *Industry Week*.

[5] ESP8266 Documentation, https://www.esp8266.com/

[6] Arduino IDE Documentation, https://www.arduino.cc/en/Guide/HomePage

[7] ESP8266 Wi-Fi Library, https://github.com/esp8266/Arduino

[8] Madakam, S., Ramaswamy, R., & Tripathi, S. (2015). "The Internet of Things (IoT): A Literature Review." *Journal of Computer and Communications*, 3(5), 164-173.